Supplementary Material for CLARA: Clip-Level Multimodal Alignment with VLM-Derived Rationales for Hateful Video Detection

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Abstract

This is the supplementary material for CLARA: Clip-Level Multimodal Alignment with VLM-Derived Rationales for Hateful Video Detection. We provide implement details and extra results analysis for CLARA that are not included in the manuscript due to the page limit.

Keywords

Hateful Video Detection, Multimodal Alignment, Mixture of Experts, Contrastive Learning, Video Transformer

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1 Prompt Design for Video-level VLM Rationale Generation

In this section, we provide additional details on how the video-level rationale RA_j is generated in CLARA. Specifically, we describe the multimodal inputs provided to the VLM, the two-step prompting strategy, the exact prompts used in our experiments, and the structured generation procedure for obtaining the final rationale used in video-level encoding.

For each video V_j, we provide the VLM with two sources of information: (i) 20 uniformly sampled frames in temporal order, (ii) the video title and full transcription (If the video title is unavailable, we only provide the transcription). We use Qwen3-VL-8B-Instruct [1] as the VLM rationale generator with a two-step prompting strategy. In the first step, the VLM is asked to produce an objective analysis of the video. In the second step, the VLM performs hateful content verification conditioned on both the original multimodal inputs and the objective analysis in the first step.

1.1 Step 1: Objective Analysis

In the first step, the VLM is prompted to describe the video content in a neutral and evidence-grounded way. The goal of this step is to obtain an objective analysis OA_j that summarizes the visual content, textual content, the overall content summary, and their cross-modal relationship, without making any hateful content judgment.

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For each video, the prompt specifies the textual inputs explicitly, whereas the visual evidence is supplied separately as image inputs to the VLM. The exact prompt used in this step is shown in Table 1.

Table 1: Prompt used in Step 1 for objective video analysis.

STEP 1 PROMPT (Objective Analysis)
 # Role
You are a professional video content verifier. Your task is to accurately summarize what the video shows and what the accompanying text conveys in an objective and neutral way.
 # Instructions
You may use:
(1) 20 frames sampled uniformly in temporal order.
(2) Video title (if available) and full transcription.
You need to:
(1) Describe what is visible in the frames.
(2) Summarize the key messages conveyed by the text.
(3) Provide an overall objective summary combining visual and textual evidence.
(4) Describe how the visuals and the text relate.
Return ONLY the following tagged format (no extra text before or after). Use the tag names exactly:
 # Output Format
VISUAL_DESCRIPTION: Neutral description of visible scenes, objects, actions, and symbols. Avoid guessing details that are not supported by the frames.
TEXTUAL_DESCRIPTION: Neutral summary of the title and transcription. Avoid guessing details that are not supported by the text.
CONTENT_SUMMARY: Overall summary combining visual and textual evidence.
CROSS_MODAL_RELATION: One label from {aligned, complementary, conflicting, unclear}.
CROSS_MODAL_EXPLANATION: Briefly explain why that label fits, focusing on whether the visuals support, add context to, contradict, or do not clarify the text evidence.
CONTEXT_ELEMENTS: List of key contextual elements that help interpret the content (e.g., entities, groups, symbols).
 # Textual Input
VIDEO_TITLE: {*video_title*}
TRANSCRIPTION: {*transcription*}

1.2 Step 2: Hateful Content Verification

In the second step, the VLM is prompted to perform hateful content verification based on both the original multimodal inputs and the objective analysis OA_j generated in the first step. This step asks the VLM to make the final judgment and provide evidence-based reasons. Compared with directly prompting for hateful or non-hateful classification in a single step, this design encourages the VLM to

Table 2: Prompt used in Step 2 for hateful content verification.

STEP 2 PROMPT (Hateful Content Verification)

 # Role

You are a professional content verifier conducting hateful content analysis. Your task is to make a careful, evidence-based judgment using the information provided.

 # Instructions

You may use:
(1) 20 frames sampled uniformly in temporal order.
(2) video title and full transcription.
(3) The objective analysis of the video provided below.

You need to:
(1) Decide whether the video content is hateful or non-hateful.
(2) If hateful, indicate whether it is explicit or implicit.
(3) List the main reasons that support your decision, focusing on observable or stated evidence rather than speculation.

Return ONLY the following tagged format (no extra text before or after). Use the tag names exactly:

 # Output Format

LABEL: <hate | non-hate>
EXPLICITNESS: <explicit | implicit | na>
CONFIDENCE: <high|medium|low>
REASONS: <reason 1>, <reason 2>, ...
NOTES: <any brief clarifications or assumptions, if needed.>

 # Textual Information

VIDEO_TITLE: {*video_title*}
TRANSCRIPTION: {*transcription*}
OBJECTIVE ANALYSIS: {*objective_analysis*}

first establish grounded multimodal understanding before making a decision. The exact prompt used in this step is shown in Table 2.

1.3 Rationale Representation Generation

After these two steps, we construct the final rationale RA_j as a structured list of strings: RA_j = [*Content summary, Visual description, Textual description, Cross-modal relation, Contextual elements, Final decision, Reasons, Notes*]. Empty fields are preserved as empty strings so that all videos share the same rationale schema. Each element and its description in RA_j are detailed in Table 3.

The resulting structured rationale RA_j is then encoded using BERT [2] to obtain its embedding E_j^{RA}:

$$E_j^{RA} = \mathrm{BERT}(RA_j) \tag{1}$$

2 Statistical Significance Analysis

To further verify that the performance gains of CLARA are not due to random variation across folds, we conduct additional pairwise t-tests at a 95% confidence level ($\alpha = 0.05$). Specifically, we compare CLARA against three strong baselines, namely MoRE, HVGuard, and MM-HSD, on four evaluation metrics: Accuracy (Acc), Macro-F1 (M-F1), Macro-Precision (M-P), and Macro-Recall (M-R). The tests are conducted on the 5-fold cross-validation results for each dataset.

Table 3: Elements of textual rationale.

Element	Description
Content summary	The CONTENT_SUMMARY from Step 1.
Visual description	The VISUAL_DESCRIPTION from Step 1.
Textual description	The TEXTUAL_DESCRIPTION from Step 1.
Cross-modal relation	A sentence combining CROSS_MODAL_RELATION and CROSS_MODAL_EXPLANATION from Step 1 in the form "*The relation between textual and visual content is {*CROSS_MODAL_RELATION*} and {*CROSS_MODAL_EXPLANATION*}*". If only one is available, we use it directly.
Contextual elements	If CONTEXTUALLY_ELEMENTS from Step 1 is a non-empty list, we convert it into a sentence of the form "*Contextually elements include: $e_1, e_2, \ldots, e_k$*", where e_i are the elements in CONTEXTUALLY_ELEMENTS. Otherwise, this field is set to an empty string.
Final decision	A sentence constructed from LABEL, EXPLICITNESS, and CONFIDENCE from Step 2. If the LABEL is non-hate, the sentence "*The video is considered non-hateful with {*CONFIDENCE*} confidence.*" If the label is hate, the sentence is "*The video is considered hateful ({*EXPLICITNESS*}) with {*CONFIDENCE*} confidence.*"
Reasons	The REASONS from Step 2.
Notes	The NOTES from Step 2.

Table 4: Paired t-test p-values comparing CLARA with strong baselines across all datasets.

Dataset	Metric	CLARA–MoRE	CLARA–HVGuard	CLARA–MM-HSD
HateMM	Acc	4.00e-3	1.63e-2	6.50e-3
	M-F1	6.10e-3	2.42e-2	7.50e-3
	M-P	2.80e-3	2.42e-2	1.34e-2
	M-R	9.40e-3	3.19e-2	1.17e-2
MHC_CN	Acc	6.00e-4	6.50e-3	1.40e-3
	M-F1	1.64e-2	3.00e-4	1.24e-2
	M-P	3.94e-2	1.62e-2	4.10e-3
	M-R	8.70e-3	1.00e-4	2.92e-2
MHC_EN	Acc	5.10e-3	1.48e-2	9.00e-3
	M-F1	7.00e-4	1.40e-3	2.19e-2
	M-P	1.05e-2	7.60e-3	1.49e-2
	M-R	1.00e-4	9.00e-4	3.77e-2
DeHate	Acc	1.77e-2	2.50e-3	8.40e-3
	M-F1	1.56e-2	2.21e-2	8.20e-3
	M-P	3.80e-3	6.80e-3	1.44e-2
	M-R	1.55e-2	2.38e-2	2.47e-2

As shown in Table 4, CLARA achieves statistically significant improvements over all three baselines across all datasets and all four evaluation metrics, with every p-value remaining below the 0.05 significance threshold. Beyond the overall significance, Table 4 also reveals several dataset-specific patterns.

First, on HateMM, all comparisons are statistically significant and the corresponding p-values remain consistently small across metrics and baselines, indicating stable improvements of CLARA over all competitors. Second, MHC_CN shows the largest variation

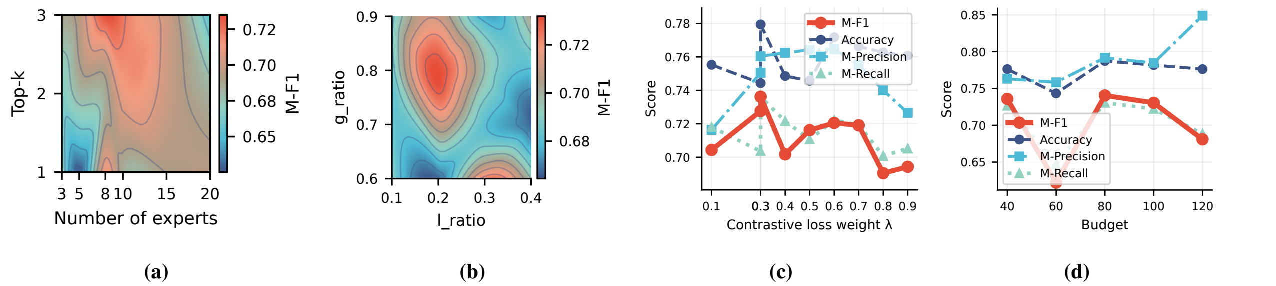


Figure 1: Parameter study of CLARA on MHC_CN. (a) the effect of the MoE configuration, including the number of experts and top-k routing, (b) the effect of the local and global temporal ratios used in contrastive learning, (c) the effect of the contrastive learning weight λ_{CL}, and (d) the effect of the training budget.

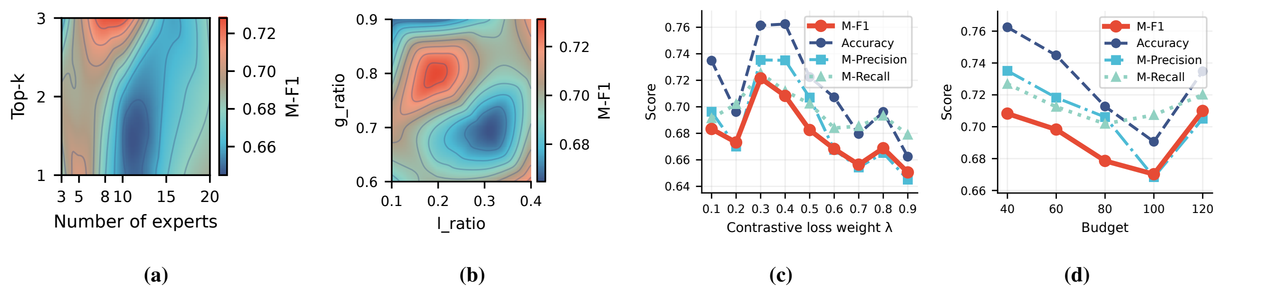


Figure 2: Parameter study of CLARA on MHC_EN. (a) the effect of the MoE configuration, including the number of experts and top-k routing, (b) the effect of the local and global temporal ratios used in contrastive learning, (c) the effect of the contrastive learning weight λ_{CL}, and (d) the effect of the training budget.

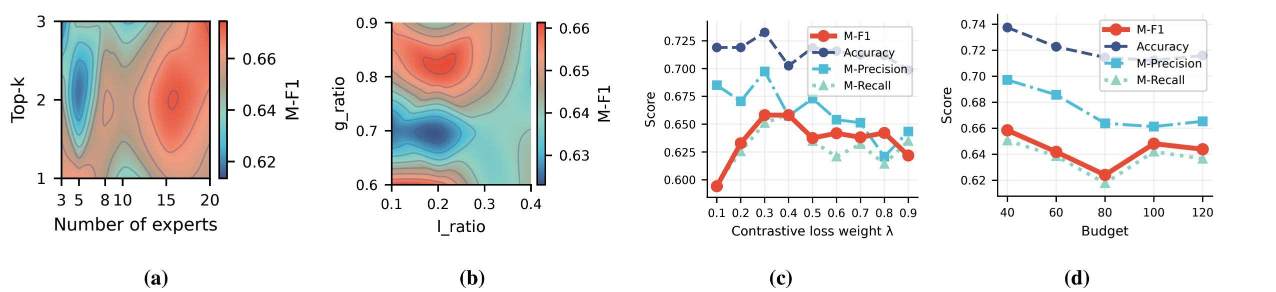


Figure 3: Parameter study of CLARA on DeHate. (a) the effect of the MoE configuration, including the number of experts and top-k routing, (b) the effect of the local and global temporal ratios used in contrastive learning, (c) the effect of the contrastive learning weight λ_{CL}, and (d) the effect of the training budget.

across baselines and metrics. While all comparisons remain significant, the strength of significance differs substantially: some comparisons yield very small p-values (e.g., against HVGuard on M-F1 and M-R), whereas others are closer to the significance threshold (e.g., against MoRE on M-P). Third, MHC_EN presents a relatively consistent pattern, with most p-values remaining below 0.01 or close to that range, although a few comparisons, especially on M-R against MM-HSD, are closer to the threshold. Finally, on DeHate, all comparisons also remain significant, with p-values distributed in a relatively narrow range. This suggests that the gains of CLARA on DeHate are statistically robust and more uniform across metrics, even though they are not as sharply separated as some of the strongest cases observed on HateMM or MHC_CN.

Overall, these results further confirm that the improvements of CLARA are statistically reliable and consistently observed across different folds and datasets.

3 Parameter Analysis on MultihateClip and DeHate

3.1 MoE Configuration

We first examine the effect of the MoE configuration on MHC_CN, MHC_EN, and DeHate by varying the number of experts and the top-k routing strategy. As shown in Figures 1a, 2a, and 3a, a broadly consistent trend can be observed across all three datasets. CLARA performs best with a moderate MoE capacity rather than with overly small or overly large configurations. On MHC_CN, the most favorable region appears when the expert pool is kept at a medium scale, between 8 and 15 experts, together with a more flexible routing strategy using top-2 or top-3 selection. MHC_EN shows a similar pattern, with the strongest performance achieved when the number of experts remains at a moderate level, around 8 to 10. On DeHate, the optimal region shifts slightly toward a somewhat larger expert pool, while top-2 routing remains the most stable choice.

Overall, these results are generally consistent with the observation on HateMM. Increasing the number of experts from a very small setting improves the model ability to capture diverse multimodal patterns, suggesting that a certain degree of expert specialization is indeed beneficial. However, further enlarging the expert pool does not continuously improve performance and may even reduce it, likely because overly sparse or fragmented routing weakens training stability and makes specialization less effective.

3.2 Local-global Ratio

We further analyze the influence of the local and global temporal ratios used in contrastive learning. As illustrated in Figures 1b, 2b and 3b, the three datasets exhibit a similar overall trend, since the best performance is obtained in an intermediate region of the ratio space. For both MHC_CN and MHC_EN, the highest M-F1 values are achieved when the local ratio is 0.2 and the global ratio is set in a relatively high range from 0.7 to 0.8. DeHate follows a similar pattern, with the most favorable region also appearing when the local ratio is kept small at 0.1 to 0.2 and the global context remains relatively large at 0.8 or above.

This result indicates that effective hateful video detection requires both short-span cues and broader contextual evidence, as observed on HateMM. The local branch helps preserve fine-grained signals that may only appear in a small number of clips, whereas the global branch provides the wider temporal context needed for interpretation. Although the exact optimum differs slightly across datasets, the general pattern remains stable. It can be observed that performance drops when the temporal design becomes extremely imbalanced, indicating that CLARA works best when local evidence and longer-range context are jointly modeled rather than overemphasizing only one temporal scale.

3.3 Contrastive Learning Weight

Next, we study the effect of the contrastive learning weight. As shown in Figures 1c, 2c, and 3c, all three datasets consistently favor a moderate contrastive weight, with the most effective range lying between 0.3 and 0.4. On MHC_CN, M-F1 reaches its highest value at 0.3. A similar pattern can be observed on MHC_EN, where the best results also appear at 0.3, followed by a gradual decline as the contrastive objective becomes overly dominant. On DeHate, M-F1 increases clearly from smaller weights, reaches its peak at 0.3, and then decreases again as the weight continues to grow.

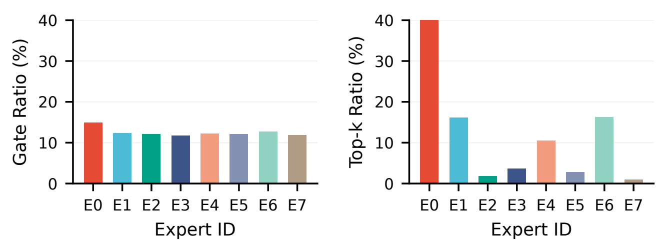
These results are consistent with the trend observed on HateMM. Introducing contrastive learning is clearly beneficial, as it helps align local and global temporal representations and improves the quality of multimodal feature learning. However, assigning too much weight to the contrastive objective can harm the final classification performance, because optimization becomes overly focused on representation alignment at the expense of the main discriminative objective. Therefore, a proper contrastive weight provides an effective trade-off between temporal alignment and hate classification.

3.4 Total Frame

Finally, we evaluate the effect of the total frame budget. As shown in Figures 1d, 2d and 3d. While the specific strengths highlighted by these three datasets vary slightly, the overall conclusion remains consistent with that of HateMM. CLARA does not continue to benefit from an infinitely increasing frame budget. On MHC_CN, performance improves from smaller budgets to a moderate budget, with the best M-F1 observed at 80 frames, while a further increase to 120 frames does not bring additional gains and instead leads to a noticeable drop in M-F1. On MHC_EN, the strongest performance is already obtained with a relatively compact budget, and increasing the number of frames generally reduces the overall effectiveness. DeHate shows a similar trend, where smaller budgets are already sufficient to achieve the best or near-best M-F1, and larger budgets tend to introduce diminishing returns.

These results show a similar trend to that observed on HateMM. Once the key multimodal evidence has already been captured, adding more frames does not necessarily improve representation quality, and may instead introduce less informative content. This suggests that CLARA is able to identify useful hateful cues with a relatively compact temporal budget, while overly large inputs may dilute the most salient signals and make optimization less effective.

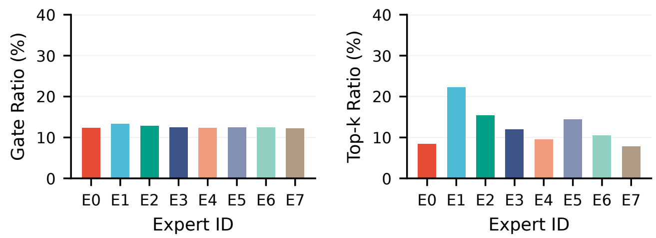
4 MoE Routing Analysis on MultihateClip and DeHate



(a) (b)

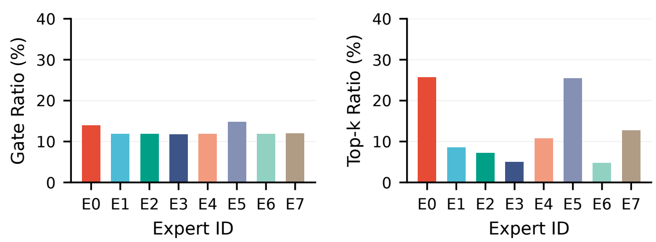
Figure 4: Expert routing statistics of the MoE on MHC_CN : (a) the distribution measured by gate probability ratio. (b) the distribution of experts measured by top-k selection ratio.

We further analyze the routing behavior of the MoE on MHC_CN, MHC_EN, and DeHate from two complementary perspectives: the gate probability ratio and the top-k selection ratio. As shown in Figures 4–6, a consistent pattern can be observed across all three datasets. The gate probability distributions are relatively balanced across experts, while the top-k selection distributions are much more uneven. This indicates that the gating network does not collapse to a dominant expert at the scoring stage. Instead, it maintains a broadly distributed preference over the expert pool, and the final sparse routing emerges mainly after top-k selection. Such a discrepancy between soft gate scores and hard expert activation suggests



(a) (b)

Figure 5: Expert routing statistics of the MoE on MHC_EN : (a) the distribution measured by gate probability ratio. (b) the distribution of experts measured by top-k selection ratio.



(a) (b)

Figure 6: Expert routing statistics of the MoE on DeHate : (a) the distribution measured by gate probability ratio. (b) the distribution of experts measured by top-k selection ratio.

that CLARA learns meaningful expert specialization rather than relying on uniformly shared routing.

Across the three datasets, the gate probability ratios remain relatively balanced across experts, indicating that all experts are retained as active candidates during routing. In contrast, the top-k selection ratios are much more uneven, suggesting that the final routing decisions rely on a smaller subset of experts. MHC_CN shows the strongest concentration pattern, where one expert dominates and only a few others are selected as secondary choices. MHC_EN exhibits a more distributed routing structure, with several experts selected at comparable frequencies and no single expert clearly dominating. DeHate lies between these two cases, with two experts emerging as the main routing destinations while the remaining experts are activated much less frequently. These results suggest that although the gating network preserves broad routing flexibility, the final sparse routing still learns dataset-specific expert specialization patterns.

Overall, the routing statistics on all three datasets consistently support the design motivation of the MoE module in CLARA. The relatively balanced gate probability ratios show that the routing network preserves the potential utility of different experts, whereas the more skewed top-k ratios confirm that the model can still make selective and sparse expert assignments when forming final representations.

Table 5: Impact of text encoder on CLARA across all datasets.

Dataset	Text Encoder	Acc	M-F1	M-P	M-R
HateMM	qwen_0.6	0.864	0.855	0.870	0.846
	qwen_8	0.878	0.871	**0.882**	0.864
	bert	**0.879**	**0.872**	0.874	**0.872**
MHC_CN	qwen_0.6	**0.780**	**0.736**	**0.767**	**0.730**
	qwen_8	0.772	0.729	0.764	0.722
	bert	0.779	0.734	0.760	0.726
MHC_EN	qwen_0.6	0.764	0.726	0.732	0.718
	qwen_8	**0.778**	0.721	**0.733**	**0.725**
	bert	0.763	**0.727**	0.732	0.723
DeHate	qwen_0.6	**0.741**	**0.662**	**0.704**	**0.652**
	qwen_8	0.730	0.637	0.692	0.629
	bert	0.735	0.659	0.697	0.650

5 Impact of Model Components: Text Encoder and Rationale Generator

5.1 Impact of Text Encoder Variants

We evaluate the impact of different text encoder choices on CLARA by comparing three variants: Qwen-based encoders [4] with embedding dimensions of 1024 (qwen_0.6) and 4096 (qwen_8), as well as a BERT-based encoder. The results are reported in Table 5.

Overall, CLARA demonstrates strong robustness across all text encoder variants, with consistently competitive performance on all datasets. Despite the substantial difference in embedding dimensionality between qwen_0.6 (1024) and qwen_8 (4096), the performance gap remains relatively small across all evaluation metrics. This indicates that the proposed framework is not sensitive to the specific representation scale of the textual features. Moreover, while BERT achieves slightly better or comparable results on some datasets (e.g., HateMM), the improvements are marginal. In contrast, Qwen-based encoders, even with significantly different embedding sizes, maintain competitive performance across all datasets. This further highlights the flexibility of CLARA in accommodating heterogeneous text representations without requiring architecture-specific tuning.

These results suggest that CLARA is highly compatible with a wide range of text encoders, including those with different model architectures and embedding dimensionalities. Such compatibility is particularly important in multimodal settings, where textual representations may originate from diverse large language models.

5.2 Impact of Rationale Generator Variants

We further investigate the impact of different VLM-based rationale generators by comparing LLaVA1.5-7B [3] and Qwen3VL-8B-Instruct [1], as shown in Table 6.

It is worth noting that the two VLMs already exhibit a clear performance gap when directly applied to the hateful video detection task. As shown in Table 2 of the main manuscript, Qwen3VL consistently achieves markedly better performance than LLaVA across all datasets and all four evaluation metrics. This indicates that Qwen3VL provides stronger multimodal reasoning and alignment capabilities at the model level. When integrated into CLARA as rationale generators, a similar trend is observed, where Qwen3VL-based rationales

Table 6: Impact of rationale generator on CLARA across all datasets.

Dataset	VLM	Acc	M-F1	M-P	M-R
HateMM	LLaVA	0.8575	0.8443	0.8577	0.8379
	Qwen3VL	**0.8790**	**0.8720**	**0.8740**	**0.8720**
MHC_CN	LLaVA	0.7084	0.6673	0.6715	0.6514
	Qwen3VL	**0.7790**	**0.7340**	**0.7600**	**0.7260**
MHC_EN	LLaVA	0.7372	0.6898	0.7067	0.6859
	Qwen3VL	**0.7630**	**0.7270**	**0.7320**	**0.7230**
DeHate	LLaVA	0.7063	0.6103	0.6515	0.5979
	Qwen3VL	**0.7350**	**0.6590**	**0.6970**	**0.6500**

consistently lead to better performance across all datasets. This suggests that higher-quality multimodal reasoning from the VLM translates into more informative and discriminative rationale representations, which further benefit downstream classification.

Overall, these results show that rationale quality has a clear impact on the performance of CLARA, with stronger VLMs producing more informative rationales and leading to better downstream results. Nevertheless, even when using the weaker LLaVA, CLARA still remains competitive against strong baselines, suggesting that the framework can effectively leverage rationales of varying quality.

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